Numpy 2 Hacker Rank

1. **Task**

Your task is to print an array of size X with its main diagonal elements as 's and 's everywhere else.

Note

In order to get alignment correct, please insert the line

np.set_printoptions(legacy='1.13')

below the numpy import.

Input Format

A single line containing the space separated values of and .
N denotes the rows.
M denotes the columns.

Output Format

Print the desired N X Marray.

Sample Input

3 3

Sample Output

```
[[ 1.  0.  0.]
 [ 0.  1.  0.]
 [ 0.  0.  1.]]
```

Change Theme Language Pypy 3

```python
# Enter your code here. Read input from STDIN. Print output to STDOUT
import numpy as np

n,m=map(int,input().strip().split())
np.set_printoptions(legacy='1.13')
print(np.eye(n,m))
```

Python ★★★★

You have earned 20.00 points!

You are now 125 points away from the gold level for your python badge.

31% 275/400

Congratulations

You solved this challenge. Would you like to challenge your friends?

Next Challenge

Test case 0

Test case 1

Test case 2

Compiler Message

```
Success
```

Input (stdin) Download

```
3 3
```

Expected Output Download

```
[[ 1.  0.  0.]
 [ 0.  1.  0.]
 [ 0.  0.  1.]]
```

2.

Task

You are given two integer arrays, A and B of dimensions NXM.

Your task is to perform the following operations:

1. Add (A + B)
2. Subtract (A - B)
3. Multiply (A * B)
4. Integer Division (A / B)
5. Mod (A % B)
6. Power (A ** B)

Note

There is a method numpy.floor_divide() that works like numpy.divide() except it performs a floor division.

Input Format

The first line contains two space separated integers, N and M.

The next N lines contains M space separated integers of array A.

The following N lines contains M space separated integers of array B.

Output Format

Print the result of each operation in the given order under **Task.**

Change Theme Language Pypy 3

```
# Enter your code here. Read input from STDIN. Print output to STDOUT
import numpy as np
n,m =map(int,input().strip().split())
arr =[]
arr1 =[]
for _ in range(n):
    arr.append(list(map(int,input().strip().split(' '))))
for _ in range(n):
    arr1.append(list(map(int,input().strip().split(' '))))

a = np.array(arr)
b = np.array(arr1)

print(a+b)
print(a-b)
print(a*b)
print(a//b)
print(a%b)
print(a**b)
```

You have earned 20.00 points!

You are now 105 points away from the gold level for your python badge.

42% 295/400

Congratulations

You solved this challenge. Would you like to challenge your friends?

Next Challenge

Test case 0

Test case 1

Test case 2

Compiler Message

```
Success
```

Input (stdin) Download

```
1 4
1 2 3 4
5 6 7 8
```

Expected Output Download

```
[[ 6  8 10 12]]
[[-4 -4 -4 -4]]
[[ 5 12 21 32]]
```

3.

Task

You are given a 1-D array, A. Your task is to print the $floor$, $ceil$ and $rint$ of all the elements of A.

Note

In order to get the correct output format, add the line

$\mathrm{numpy.set\_printoptions(legacy='1.13')}$ below the numpy import.

Input Format

A single line of input containing the space separated elements of array A.

Output Format

On the first line, print the $floor$ of A.

On the second line, print the $ceil$ of A.

On the third line, print the $rint$ of A.

Sample Input

```
1.1 2.2 3.3 4.4 5.5 6.6 7.7 8.8 9.9
```

Sample Output

```
[ 1.  2.  3.  4.  5.  6.  7.  8.  9.]
[  2.   3.   4.   5.   6.   7.   8.   9.  10.]
[  1.   2.   3.   4.   6.   7.   8.   9.  10.]
```

```python
# Enter your code here. Read input from STDIN. Print output to STDOUT
import numpy as np

np.set_printoptions(legacy='1.13')
arr =np.array(list(map(float,input().strip().split())))

print(np.floor(arr))
print(np.ceil(arr))
print(np.rint(arr))
```

Python ★★★★

You have earned 20.00 points!

You are now 85 points away from the gold level for your python badge.

53% 315/400

Congratulations

You solved this challenge. Would you like to challenge your friends?

Next Challenge

Test case 0

Test case 1

Test case 2

Compiler Message

```
Success
```

Input (stdin) Download

```
1.1 2.2 3.3 4.4 5.5 6.6 7.7 8.8 9.9
```

Expected Output Download

```
[ 1.  2.  3.  4.  5.  6.  7.  8.  9.]
[  2.   3.   4.   5.   6.   7.   8.   9.  10.]
[  1.   2.   3.   4.   6.   7.   8.   9.  10.]
```